

Course: Databases — Laboratory Work 1

Topic: Entity-Relationship Diagram (ERD) Solution

Problem Description: A university database system managing Students, Instructors, Courses, and Course Registrations.

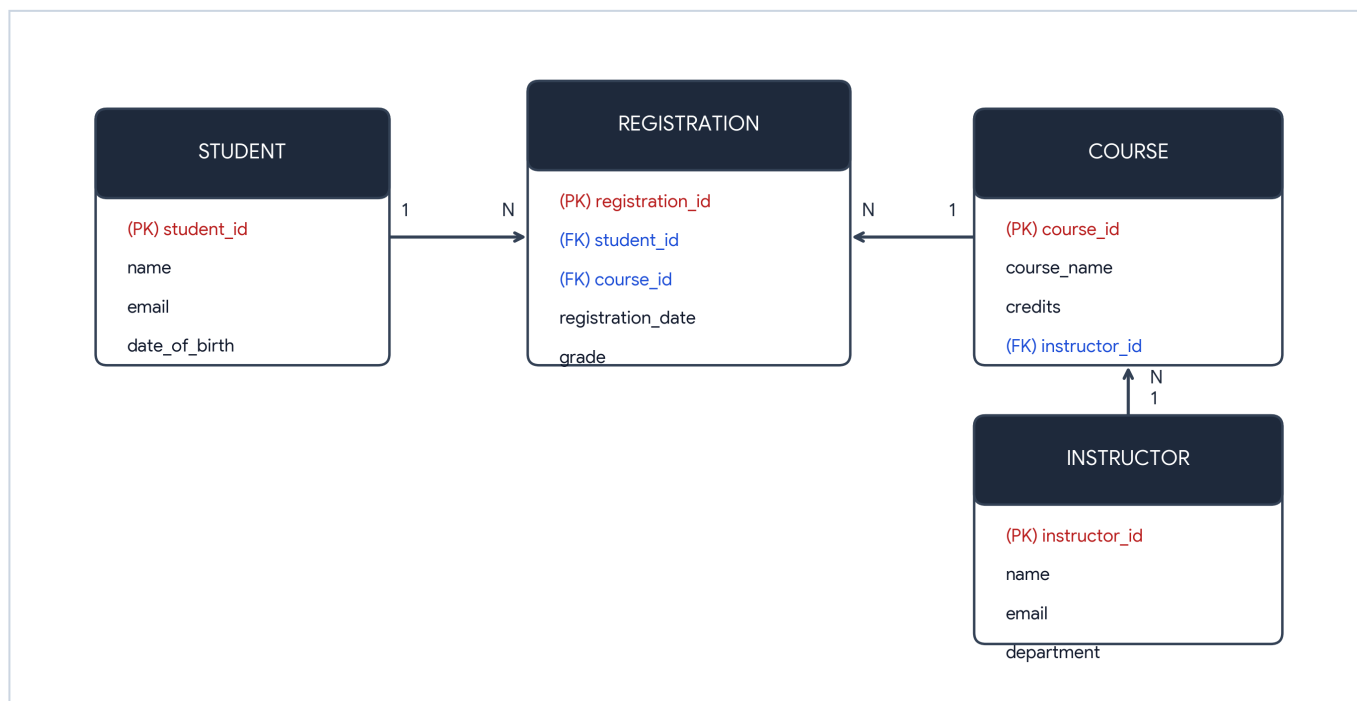
1. Entity & Attribute Table

Entity Name	Attribute Name	Data Type	Key / Constraint	Description
Student	student_id	INT	PK (Primary Key)	Unique ID for student
	name	VARCHAR(100)	Required	Student full name
	email	VARCHAR(100)	Unique, Required	Student email address
	date_of_birth	DATE	Required	Date of birth
Instructor	instructor_id	INT	PK (Primary Key)	Unique ID for instructor
	name	VARCHAR(100)	Required	Instructor full name
	email	VARCHAR(100)	Unique, Required	Instructor email address
	department	VARCHAR(100)	Required	Department name
Course	course_id	VARCHAR(10)	PK (Primary Key)	Unique course code
	course_name	VARCHAR(100)	Required	Course title
	credits	INT	Required	Number of credits
	instructor_id	INT	FK (Foreign Key)	Links to Instructor table
Registration (Associative Entity)	registration_id	INT	PK (Primary Key)	Unique registration ID
	student_id	INT	FK (Foreign Key)	Links to Student table
	course_id	VARCHAR(10)	FK (Foreign Key)	Links to Course table
	registration_date	DATE	Required	Date of course enrollment
	grade	VARCHAR(2)	Optional	Final grade earned

2. Relationships & Cardinality

- **Instructor to Course (1 : N):** One instructor can teach many courses. Each course has exactly one instructor.
- **Student to Course (M : N via Registration):** A student can register for many courses, and a course can have many students. This many-to-many relationship is split using the Registration entity.

3. Entity-Relationship Diagram (ERD)



4. Legend & Solution Description

- **Rectangle Boxes**: Represent entities (database tables).
- **PK (Primary Key)**: Unique identifier for each table row.
- **FK (Foreign Key)**: Link to another table's Primary Key.
- **1 : N Arrows**: Shows a One-to-Many relationship direction.
- **Summary**: The database structure accurately stores students, instructors, courses, and course enrollments while enforcing correct keys and constraints.